

Pricing Based Charging Navigation Scheme for Highway Transportation to Enhance Renewable Generation Integration

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Abstract—The charging operation is rapidly becoming a key concern for highway charging station operation as the electric vehicle charging demand for highway traveling rises. This paper proposes a charging price-setting scheme for highway charging station operation, considering distributed renewable energy generators neighboring highway charging stations. For this purpose, a bi-level approach is proposed in this paper to model the conflicting objectives between the highway charging station operator and electric vehicles to optimize charging price and their routes. The upper level is designed to optimize the highway charging station operation's profit. The lower level characterizes the charging behavior of EVs via a time-space network model. To solve the proposed bi-level model, we adopt the column-and-constraint generation algorithm to obtain an optimal result. The performance of the proposed scheme is illustrated by numerical case studies based on an experimental highway system.

Index Terms—Bi-level mixed integer programming, charging price-setting scheme, electric vehicle, highway transportation, renewable energy utilization.

NOMENCLATURE

Indices and Sets

A_x	Set of the traveling arcs.
A_y	Set of the charging arcs.
N	Set of nodes in time-space network.
N^{start}	Set of start nodes in time-space network.

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N^{end}	Set of end nodes in time-space network.
$N^{virtual}$	Set of virtual nodes in time-space network.
$N^{station}$	Set of charging nodes in time-space network.
j	Index of enumerating integer variables.
m	Index of electric vehicle charging station.
p	Index of feasible routes.
t	Index of time.
φ	Index of charging price option.
ω	Index of electric vehicle.

Parameters

E_{ω}^{start}	Power at the start of the point for electric vehicle ω .
E_{ω}^{end}	Power at the end of the point for electric vehicle ω .
E_{ω}^{max}	Maximum battery capacity for electric vehicle ω .
E_{ω}^{min}	Minimum battery capacity for electric vehicle ω .
M	Number of charging stations.
N	Number of nodes in time-space network.
$P_{m,wind}^t$	Wind power generation at m charging station in time interval t .
$P_{m,pv}^t$	Photovoltaic generation at m charging station in time interval t .
$P_{m,max}$	Maximal charging power limit at m charging station.
$P_{m,t}^{grid}$	Purchased power from the main grid.
P	Set of feasible routes in time-space network.
T	Number of time intervals.
W_{max}	Maximum revenue from the sale of electricity.
x_0	Represents if the vehicle is departing at the point.
θ	The optimal solution to the sub-problem.
ξ^+	Charging rate in a time interval.
ξ^-	Energy consumption rate in a time interval.
$\pi_{m,t,\varphi}^{HCSO}$	Charging price option φ at m charging station in time interval t .
$\pi_{m,t}^{grid}$	The main grid electricity price at m charging station in time interval t .
Φ	Number of Electricity price options.
Ω	Number of electric vehicles.
ℓ	The optimal solution to the master problem.

Decision Variables

$I_{m,t,\varphi}^{HCSO}$	Binary variable indicating whether to adopt price $\pi_{m,t,\varphi}^{HCSO}$.
$P_{m,t}^{HCSO}$	Total charging load at station m in time interval t .

$W_{m,t,\varphi}^{\text{HCSO}}$	Total charging load at price $\pi_{m,t,\varphi}^{\text{HCSO}}$.
$x_{n,t}^{\omega}$	Binary variable indicating whether electric vehicle ω choose this traveling arc.
$y_{n,t}^{\omega}$	Binary variable indicating whether electric vehicle ω choose this charging arc.
μ	Dual variables for lower level model constraints.

I. INTRODUCTION

CONSIDERING the ever-growing threats of environmental pollution and climate change [1], [2], [3], electric vehicles (EVs) have attracted an increasing amount of attention. Meanwhile, the market scale of the EV is gradually expanding under the improved vehicle battery [4] technology and supporting policies. However, one of the most challenging application scenarios for EVs is highway transportation considering the limited driving range due to the battery technology. To this end, it requires economic and efficient charging scheduling for EVs in the highway transportation scenario.

In particular, the recent development of distributed energy resources, such as highway roadside photovoltaic (PV) resources [5], [6], [7] and distributed renewable generation based on service zone [8], make it possible to integrate more renewable energy into the highway transportation sector. However, it presents multiple challenges to efficiently utilize those clean energy resources due to the intermitted nature of generation profile and uncoordinated arrival of charging demand. Intuitively, strategically managing the charging demand side for highway traffics and matching it with renewable generation resources become a potential solution to those challenges.

The charging demand management problem in the context of highway transportation has been investigated by numbers of researchers. Among those research works, most of them focus on transportation methods [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. Del Razo et al. [9] proposed a distributed scheduling approach for coordinating charging stops planned on a highway trip on the basis of the A* algorithm while the objective is to reduce the total traveling time for each EV. Yang et al. [10] combined two types of charging station (CS) searching algorithms. One only utilizes local information for an EV, while the other applies global information obtained through a central server. A charging reservation mechanism is provided for the EVs on highways in [11], improving the charging facilities utilization and meeting users' satisfaction based on traveling time flexibility and charging time. Kim et al. [12] proposed a distributed on-ramp-flow coordination scheme and a main-stream-inflow control method based on distributed consensus algorithms. The inflow and outflow of highway traffic are effectively harmonized so as to avoid road traffic congestion. Reference [13] and [14] developed a distributed scheduling algorithm to adjust the EV charging percentage at individual stations so that all the charging resources along the highway are well utilized while each EV minimizes waiting time on highways. A refreshing EV fast-charging strategy was proposed in [15] to provide simultaneous charging service for EVs waiting in line, utilizing the residual power of fast charging piles with multiple power output connections. Anderson et al.

[16] investigated the charging scheduling problems for EVs on highways from various perspectives, such as utilities, electricity providers, station operators, and other infrastructure. Rahman et al. [17] modeled the EV navigation management problem using Satisfiability Modulo Theories and presented a price-based navigation control technique to achieve better load balancing for the system. Wang et al. [18] developed a multistage optimization model and derived an intercity travel charging strategy to support EV drivers. This model determines the optimal charging energy for each CS located along the available routes with the aim of minimizing traveling time and charging costs. Hou et al. [19] proposed a simultaneous multi-round auction to solve the highway charging scheduling problem by allowing users to bid on stations, charging times, and energy simultaneously, with the aim of maximizing the total revenue of CSs. Although those aforementioned works provided considerable contribution to the efficient charging scheduling for highway transportation, the detailed characteristics of charging station operation have not been specified.

A few researchers explored the highway EV charging scheduling problem from an interdisciplinary approach which considers the coupled operation of the power system and transportation system. Zhang et al. [20] studied the EV routing problem on a coupled transportation and power network to promote the utilization of renewable energy sources, thereby effectively enhancing social welfare. Lv et al. [21] investigated the power flow problem for transmission networks with a wireless charging strategy for EVs as a game model based on locational marginal prices (LMP) interaction. However, the service zone based charging pricing scheme, which is the most commonly used charging demand management scheme in the highway transportation scenario, is not considered in those works.

Although pricing based charging demand management has not been studied in the highway transportation applications, there are numbers of works [22], [32] that explored it in the context of urban road transportation cases. Chen et al. [22] proposed a dynamic pricing vector formation model based on the K-means algorithm to extract historical price data and form a date-specific price vector. Zhang et al. [23] studied delay-optimal charging scheduling of the EVs at a CS considering grid power price. Chen et al. [24] investigated energy management policy for EV charging service providers on the basis of stochastic dynamic programming algorithms and greedy algorithms. A hierarchical game approach was proposed in [25] to attract EVs to charge during low load periods and reduce time costs in implementing vehicle navigation. Li et al. [26] proposed a price incentive-based charging navigation strategy for EVs to reduce the charging cost and relieve the traffic burden. Sun et al. [27] presented a method for a robust charging scheduling strategy of overnight charging of EVs in low voltage networks. Cui et al. [28] modeled the optimal charging pricing problem as a bi-level programming considering the interdependent operations of power system and transportation from the view of the charging network operator. Ding et al. [29] proposed a stochastic resource-planning scheme for plug-in hybrid EV charging stations to manage the associated risk. Qiu et al. [30] solved EV pricing problem by combining deep deterministic policy gradient principles with a prioritized

experience replay strategy. Aljohani et al. [31] designed a dynamic, real-time, demand-influenced energy pricing mechanism to allocate fairness to the billing strategy of EVs charging. Tao et al. [32] modeled a pricing strategy and charging management problem for PV-assisted EV charging stations based on the Stackelberg game. Although the above-mentioned works have made significant progress, there are still noticeable gaps to fill. In particular, these papers hardly characterize the large spatial span of highway traveling, which introduces unique operational characteristics for both transportations (e.g., road network topology and EV operation characteristics) and power systems (e.g., distributed renewable generation, wholesale market LMP) sides.

To address the aforementioned limitations, we propose a pricing-based charging navigation scheme for highway transportation to enhance renewable generation integration. In the proposed model, the highway charging station operator (HCSO) determines the charging pricing policy for the CSs in each service zone and energy management scheme to procure electricity from the wholesale power market while accounting the utilization of distributed renewable generation resources in the service zones. Correspondingly, each EV optimizes its trip costs while accounting for the charging costs and traveling constraints. To model this interaction among HCSO and EVs, we establish a bi-level model and adopt a time-space network formulation to characterize the operation characteristics of EVs in the highway transportation context. The main contributions of this paper are summarized as shown in the following.

- 1) We propose a pricing-based charging navigation scheme in the context of highway transportation while accounting for the spatial-temporal characteristics of distributed renewable power generation. To model the charging behavior of EVs in the highway transportation system, we develop a time-space network model and formulate it as a mixed-integer linear program (MILP).
- 2) We establish a bi-level optimization formulation for the pricing-based charging navigation model for the highway transportation system. The upper level aims to optimize the operation profit of highway charging stations and the lower level is designed to optimize the charging cost of EVs. The column-and-constraint generation (CCG) method is deployed to solve the bi-level optimization problem.

This paper is organized as follows. Section II describes the framework of the problem. Section III introduces the CCG based solution method. A numerical case study is presented in Section IV while Sections V presents conclusions of this paper.

II. MODEL DESCRIPTION

A. General Description

In this paper, we propose a charging pricing scheme for HCSO. The pricing problem is constructed as a bi-level model to obtain the EV charging scheduling and spatial-temporal pricing strategy.

An integrated operation framework is shown in Fig. 1. We assume that HCSO is a cross-regional operator with numbers of distributed renewable resources, including PV and wind

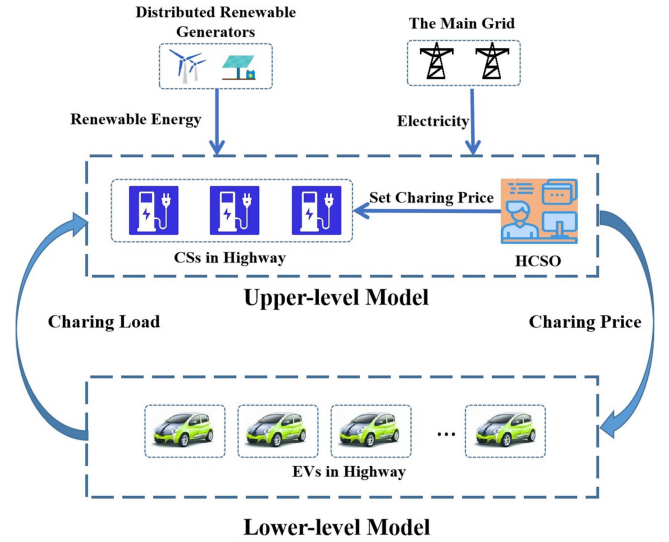


Fig. 1. Integrated framework of HCSO and EVs.

resources in its service zones. The HCSO forecasts the spatial-temporal distribution of charging demand, renewable generation, and wholesale electricity price for the next operating day. Meanwhile, we assume EVs declare their next day itinerary to HCSO, which can be conducted via mobile apps such as E-charging [33]. Based on that information, the HCSO optimizes the charging pricing strategies for each CS in the highway system to maximize its total operation profit. Based on the charging prices, each EV optimizes its routing and charging plan for the declared itinerary.

It shall be noted that the pricing scheme for HCSO needs to consider the price-responsive behavior of EVs. To characterize this interactive feature, we develop a bi-level formulation in which the upper level model depicts HCSO decision-making process while the lower level model portrays the EVs' price responsive behaviors. For a simplistic purpose, we do not consider the uncertainties in this paper. However, it can be easily incorporated into the proposed scheme by adopting well-established stochastic programming techniques [34].

B. Time-Space Network for EV Decision Making

We establish a time-space network based formulation to model EV's routing and charging decisions. Compared to the commonly adopted network model in urban road transportation [35], [36], [37], [38], the routing and charging in highway transportation are different due to its unique network topology. In particular, two distinctive features can be summarized as shown in the following.

- 1) Irreversible path: the routing decisions are one-directional. In other words, EVs cannot travel backward in the highway system.
- 2) Single routing path: there is only one routing choice for each origin and destination pair. The routing decision for each EV is choosing which service zone to stop and charge.

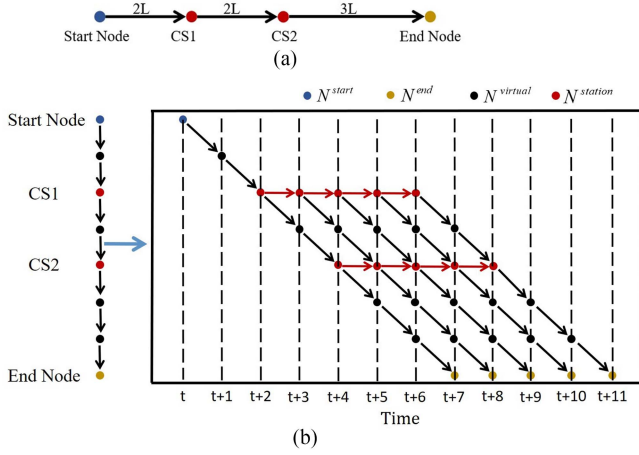


Fig. 2. Illustrative highway system. (a) single-line model (b) time-space network model.

To characterize those two features, we develop a time-space network model and illustrate it with a toy system. As shown in Fig. 2, we consider a highway with two charging stations. To characterize the spatial-temporal routing decision for a single EV, we convert the single-line model to a time-space network model. We use the red arrows to represent the charging arc and the black arrows to depict the traveling arc. In other words, we model the charging behavior of an EV by making it travel through a charging arc. Similarly, we model the routing behavior of an EV by letting it transverse a traveling arc.

Notably, the distance between two nodes adjacent to each other in the time-space network shall be the same. However, it can be observed that the traveling time between adjacent nodes in Fig. 2(a) is different. This is always the case in most real-world situations. To address this issue, we introduce virtual nodes, denoted by black nodes in Fig. 2(b), to guarantee consistent distance between each adjacent node pair. For example, if a unit of time in the time-space network is 15 minutes and it takes 1 hour to travel between CS1 and CS2, then we add three virtual nodes between those two charging stations to make sure the traveling time between each adjacent node pair are the same (i.e., 15 minutes). Also, to guarantee continuous traveling in highway transportation, we define that EV is not allowed to stop at the virtual node. In contrast to that, the EV has two options when it arrives at the charging nodes (i.e., service zones). It can either stop to charge or continue its trip. Moreover, we assume that the charging power is constant. As illustrated by Fig. 2(b), an EV can fully charge itself from minimum state of charge (SoC) to full charge state in 4-time intervals.

C. Upper Level Model

We introduce the mathematical formulation of an upper level model in this subsection. The objective of the upper level is to maximize the revenue of HCSO, as defined in (1). The first term in (1) represents the revenue from selling electricity to EVs and the detailed expression is shown in (2). The second term performs the cost of purchasing electricity from the main

grid.

$$\max \sum_{t=1}^T \sum_{\varphi=1}^{\Phi} \sum_{m=1}^M W_{m,t,\varphi}^{HCSO} - \sum_{m=1}^M \sum_{t=1}^T \pi_{m,t}^{\text{grid}} \cdot P_{m,t}^{\text{grid}} \quad (1)$$

$$W_{m,t,\varphi}^{HCSO} = \pi_{m,t,\varphi}^{HCSO} \cdot P_{m,t}^{HCSO} \cdot I_{m,t,\varphi}^{HCSO} \quad (2)$$

It should be noted that the right side of (2) is nonlinear. Therefore, we adopt the McCormick envelop to convert it to the linear form, as shown in the following constraints:

$$\begin{aligned} W_{m,t,\varphi}^{HCSO} &\geq \pi_{m,t,\varphi}^{HCSO} \cdot P_{m,\max} \cdot I_{m,t,\varphi}^{HCSO} + \pi_{m,t,\varphi}^{HCSO} \cdot P_{m,t}^{HCSO} \\ &\quad - \pi_{m,t,\varphi}^{HCSO} \cdot P_{m,\max} \\ &\quad \forall m, t, \varphi \end{aligned} \quad (3)$$

$$W_{m,t,\varphi}^{HCSO} \leq \pi_{m,t,\varphi}^{HCSO} \cdot P_{m,\max} \cdot I_{m,t,\varphi}^{HCSO} \quad \forall m, t, \varphi \quad (4)$$

$$W_{m,t,\varphi}^{HCSO} \leq \pi_{m,t,\varphi}^{HCSO} \cdot P_{m,t}^{HCSO} \quad \forall m, t, \varphi \quad (5)$$

For the charging pricing process, we assume the HCSO can determine its pricing strategy for a specific location and time by selecting from a certain number of price candidates. This is often the case in lots of charging infrastructures [17]. Therefore, we introduce a set of binary variables to indicate whether to select a certain pricing candidate for the target CS for a specific certain time, as shown in constraint (6).

$$\sum_{\varphi=1}^{\Phi} I_{m,t,\varphi}^{HCSO} = 1 \quad \forall m, t \quad (6)$$

Constraint (7) denotes the power adequacy constraints for CSs. Considering the HCSO is the monopolistic operator, we add a capping constraint for the operation profit, as shown in (8), to avoid the HCSO overcharge in the proposed charging navigation scheme.

$$P_{m,t}^{pv} + P_{m,t}^{\text{wind}} + P_{m,t}^{\text{grid}} \geq P_{m,t}^{\text{charge}} \quad \forall m, t \quad (7)$$

$$\sum_{t=1}^T \sum_{\varphi=1}^{\Phi} \sum_{m=1}^M W_{m,t,\varphi}^{HCSO} \leq W_{\max} \quad (8)$$

D. Lower Level Model

The lower level represents the operation of EVs traveling on the highway. The objective of each individual EV is to minimize its charging cost while accounts its routing constraints, as defined in (9).

$$\min \sum_{n=1}^M \sum_{\varphi=1}^{\Phi} \sum_{t=1}^T W_{m,t,\varphi}^{HCSO} \quad (9)$$

We introduce two types of binary variables to guarantee the irreversible and single-routing features for EVs in the highway transportation system. These binary variables in the time-space network are defined as shown in Fig. 3. For the point (n, t) , $x_{n-1,t-1}^{\omega}$ represents that EV ω moves towards the node n at time interval t and $x_{n,t}^{\omega}$ indicates that EV ω moves away from the point n at time interval t . If n is a CS node, the variable y is designed to depict the charging behavior of EVs. Similarly, $y_{n-1,t-1}^{\omega}$ represents that EV ω charges at this CS at time interval

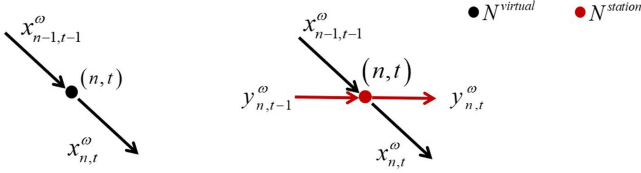


Fig. 3. Definition of binary variables in the model.

$t-1$ and $y_{n, t}^\omega$ indicates that EV ω continues to charge at time interval t .

Constraint (10) denotes the departure time of each EV. Constraints (11) and (12) ensure that EV ω must be either in a traveling arc or a charging arc in time slot t . Constraints (13) and (14) ensure the flow balance of each transportation node in the time-space network. Constraint (15) indicates that each EV arrives at its destination with the SoC level higher than the threshold value, which is determined by the traveling purpose of EVs. Constraints (16) and (17) guarantee that the vehicle charge will not exceed the limit in the time-space network. $P_{(n, t, \omega)}$ indicates the set of all paths from the departure point of the vehicle ω to point (n, t) in the time-space network. Constraint (18) restricts the maximum charging power for each CS during each time interval. Constraint (19) represents the coupling relationship between the upper and lower variables.

$$x_{n, t}^\omega = x_0 \quad \forall (n, t) \in N^{\text{start}}, \forall \omega \in \Omega \quad (10)$$

$$\sum_{n \in N^v} x_{n, t}^\omega = 1 \quad \forall t, \omega \quad (11)$$

$$\sum_{n \in N^s} (x_{n, t}^\omega + y_{n, t}^\omega) = 1 \quad \forall t, \omega \quad (12)$$

$$x_{n-1, t-1}^\omega = x_{n, t}^\omega \quad \forall (n, t) \in N^{\text{virtual}}, \forall \omega \in \Omega \quad (13)$$

$$x_{n-1, t-1}^\omega + y_{n, t-1}^\omega = x_{n, t}^\omega + y_{n, t}^\omega \quad \forall (n, t) \in N^{\text{station}}, \forall \omega \in \Omega \quad (14)$$

$$E_\omega^{\text{start}} - \xi^- \times \sum_{(n_x, t_x) \in A_x} x_{n, t}^\omega + \xi^+ \times \sum_{(n_y, t_y) \in A_y} y_{n, t}^\omega \geq E_\omega^{\text{end}} \quad \forall \omega \in \Omega \quad (15)$$

$$E_\omega^{\text{start}} - \xi^- \times \sum_{(n_x, t_x) \in P_{(n, t, \omega)}} x_{n, t}^\omega + \xi^+ \times \sum_{(n_y, t_y) \in P_{(n, t, \omega)}} y_{n, t}^\omega \geq E_\omega^{\text{min}} \quad \forall \omega \in \Omega, p \in P, (n, t) \in N \quad (16)$$

$$E_\omega^{\text{start}} - \xi^- \times \sum_{(n_x, t_x) \in P_{(n, t, \omega)}} x_{n, t}^\omega + \xi^+ \times \sum_{(n_y, t_y) \in P_{(n, t, \omega)}} y_{n, t}^\omega \leq E_\omega^{\text{max}} \quad \forall \omega \in \Omega, p \in P, (n, t) \in N \quad (17)$$

$$\xi^+ \times \sum_{\omega \in \Omega} y_{n, t}^\omega \leq P_{m, \text{max}} \quad \forall (n, t) \in N^{\text{station}} \quad (18)$$

$$P_{m, t}^{\text{HCSO}} = \sum_{\omega=1}^{\Omega} \sum_{(n_y, t_y) \in A_y} y_{n, t}^\omega \quad \forall m, t \quad (19)$$

III. SOLUTION METHOD

Since binary variables exist at both levels, the bi-level mixed integer program (BIMIP) is intractable to solve directly. We develop a CCG algorithm [39] to iteratively solve BIMIP programming effectively. This algorithm generates constraints and variables dynamically and iterates for more compact upper bounds and lower bounds to solve the mathematical model effectively. The solving process is summarized in Algorithm 1.

The master problem in Algorithm 1 relaxes the bi-level model and adds constraints at each iteration. These Karush–Kuhn–Tucker (KKT) condition based constraints compose an expanded set and provide an updated lower bound. The master problem is formulated as follows:

$$\ell = \max (1)$$

$$s.t. (3) - (8), (10) - (18)$$

$$P_{m, t}^{\text{HCSO}} \leq \sum_{\omega=1}^{\Omega} \sum_{(n_y, t_y) \in A_y} y_{n, t}^\omega \quad \forall m, t \quad (20)$$

$$P_{m, t}^{\text{HCSO}} \geq \sum_{\omega=1}^{\Omega} \sum_{(n_y, t_y) \in A_y} y_{n, t}^\omega \quad \forall m, t \quad (21)$$

$$\begin{aligned} & \sum_{n=1}^M \sum_{\varphi=1}^{\Phi} \sum_{t=1}^T W_{m, t, \varphi}^{\text{HCSO}, 0} + \alpha \left(\sum_{(n_x, t_x) \in A_x} x_{n, t}^{\omega, 0} + \sum_{(n_y, t_y) \in A_y} y_{n, t}^{\omega, 0} \right) \\ & \leq \sum_{n=1}^M \sum_{\varphi=1}^{\Phi} \sum_{t=1}^T W_{m, t, \varphi}^{\text{HCSO}} \\ & + \alpha \left(\sum_{(n_x, t_x) \in A_x} x_{n, t}^{\omega, j} + \sum_{(n_y, t_y) \in A_y} y_{n, t}^{\omega, j} \right) \quad \forall m, t, \varphi \end{aligned} \quad (22)$$

$$\mu_{m, t, 1}^j \geq 0, -\mu_{m, t, 2}^j \geq 0 \quad \forall m, t \quad (23)-(24)$$

$$-\mu_{m, t, \varphi, 3}^j \geq -1, \pi_{m, t, \varphi}^{\text{HCSO}} \mu_{m, t, \varphi, 3}^j \geq 0 \quad \forall m, t, \varphi \quad (25)-(26)$$

$$\mu_{m, t, \varphi, 4}^j \geq -1 \quad \forall m, t, \varphi \quad (27)$$

$$\mu_{m, t, \varphi, 5}^j \geq -1, -\pi_{m, t, \varphi}^{\text{HCSO}} \mu_{m, t, \varphi, 5}^j \geq 0 \quad \forall m, t, \varphi \quad (28)-(29)$$

$$P_{m, t}^{\text{charge}} \perp \mu_{m, t, 1}^j \quad \forall m, t \quad (30)$$

$$P_{m, t}^{\text{charge}} \perp \mu_{m, t, 2}^j \quad \forall m, t \quad (31)$$

$$(-\mu_{m, t, \varphi, 3}^j + 1) \perp W_{m, t, \varphi}^{\text{HCSO}} \quad \forall m, t, \varphi \quad (32)$$

$$\pi_{m, t, \varphi}^{\text{HCSO}} \mu_{m, t, \varphi, 3}^j \perp P_{m, t, \varphi}^{\text{HCSO}} \quad \forall m, t, \varphi \quad (33)$$

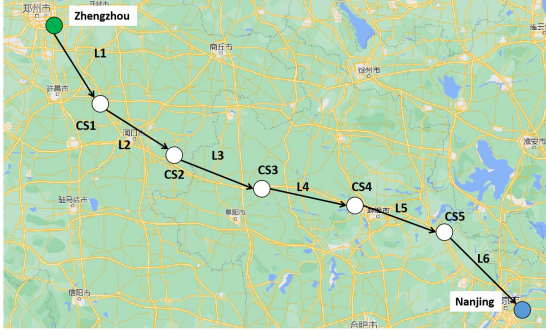


Fig. 4. Topological network in this case study.

$$W_{m,t,\varphi}^{\text{HCSO}} \perp (\mu_{m,t,\varphi,4}^j + 1) \quad \forall m, t, \varphi \quad (34)$$

$$(\mu_{m,t,\varphi,5}^j + 1) \perp W_{m,t,\varphi}^{\text{HCSO}} \quad \forall m, t, \varphi \quad (35)$$

$$-\pi_{m,t,\varphi}^{\text{HCSO}} \mu_{m,t,\varphi,5}^j \perp P_{m,t,\varphi}^{\text{HCSO}} \quad \forall m, t, \varphi \quad (36)$$

$$\left(P_{m,t}^{\text{charge},j} - \sum_{\omega=1}^{\Omega} \sum_{(n_y, t_y) \in A_y} y_{n,t}^{\omega,j} \right) \perp \mu_{m,t,1}^j \quad \forall m, t \quad (37)$$

$$\left(-P_{m,t}^{\text{charge},j} + \sum_{\omega=1}^{\Omega} \sum_{(n_y, t_y) \in A_y} y_{n,t}^{\omega,j} \right) \perp \mu_{m,t,2}^j \quad \forall m, t \quad (38)$$

$$\begin{aligned} & (-W_{m,t,\varphi}^{\text{HCSO},j} + \pi_{m,t,\varphi}^{\text{HCSO},j} \cdot P_{m,\max} \cdot I_{m,t,\varphi}^{\text{HCSO}} + \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,t}^{\text{HCSO},j} \\ & - \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,\max}) \perp \mu_{m,t,\varphi,3}^j \quad \forall m, t, \varphi \end{aligned} \quad (39)$$

$$(W_{m,t,\varphi}^{\text{HCSO},j} - \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,\max} \cdot I_{m,t,\varphi}^{\text{HCSO}}) \perp \mu_{m,t,\varphi,4}^j \quad \forall m, t, \varphi \quad (40)$$

$$(W_{m,t,\varphi}^{\text{HCSO},j} - \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,t}^{\text{HCSO},j}) \perp \mu_{m,t,\varphi,5}^j \quad \forall m, t, \varphi \quad (41)$$

To facilitate generating the dual constraints of the coupling constraint (19), we reformulate the constraint to two inequality constraints, as shown in (20)–(21). Noted that j denotes the iteration number and constraints (22)–(41) are not involved in the initial formulation. In the following iterations, constraints (22)–(41) are added to the master problem and the problem keeps iterating until convergence. Furthermore, the coupling constraints (30)–(41) are linearized by replacing $a \perp b$ with the set of linear constraints in formulation (42), with M in (42) being a large enough positive constant.

$$a \geq 0, b \geq 0, a \leq kM, b \leq (1-k)M, k \in \{0, 1\} \quad (42)$$

Based on the optimization results (e.g., charging prices in each CS) in the upper level problem, EVs optimize their routing and charging decisions to minimize charging costs. We formulate this response behavior as the sub-problem.

$$\theta = \min (9)$$

$$s.t. (5), (10 - 18), (20 - 21)$$

$$W_{m,t,\varphi}^{\text{HCSO}} \geq \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,\max} \cdot I_{m,t,\varphi}^{\text{HCSO},*} + \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,t}^{\text{HCSO}}$$

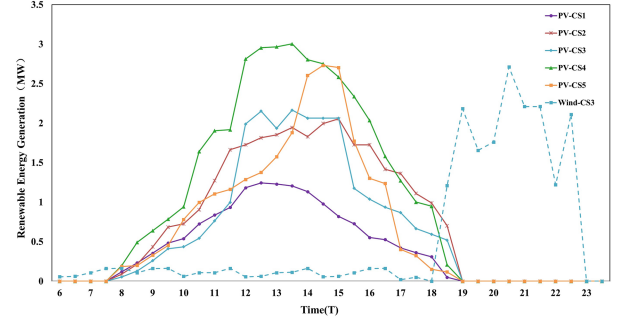


Fig. 5. Renewable generation profile for each CS.

Algorithm 1: Column-and-Constraint Generation Algorithm for BIMIP.

- 1: Initialize $UB(0) = \infty$, $LB(0) = -\infty$, and set iteration index $j = 0$;
 - 2: **while** $UB(j) - LB(j) < \varepsilon$ **do**
 - 3: **if** the master problem is infeasible **then**
 - 4: Terminate; the original optimization problem is infeasible;
 - 5: **else**
 - 6: Solve the master problem and obtain an optimal solution $(W_{m,t,\varphi}^{\text{HCSO},*}, I_{m,t,\varphi}^{\text{HCSO},*})$;
 - 7: Update $LB(j) = \zeta^*$
 - 8: **if** $UB(j) - LB(j) < \varepsilon$ **then**
 - 9: return $UB(j)$ and the corresponding solution;
 - 10: **else** solve the sub-problem for given $(W_{m,t,\varphi}^{\text{HCSO},*}, I_{m,t,\varphi}^{\text{HCSO},*})$;
 - 11: Obtain an optimal solution $(x_{n,t}^{\omega,*}, y_{n,t}^{\omega,*}, P_{m,t}^{\text{HCSO},*})$ and update $UB(j)$;
 - 12: Set $(x_{n,t}^{\omega,j+1}, y_{n,t}^{\omega,j+1}) = (x_{n,t}^{\omega,*}, y_{n,t}^{\omega,*})$, create variables $(P_{m,t}^{\text{HCSO},j+1}, \mu^{j+1})$, and add the constraints (20)–(39) to the master problem;
 - 13: **end if**
 - 14: **end if**
 - 15: **end while**
-

$$-\pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,\max} \quad \forall m, t, \varphi \quad (43)$$

$$W_{m,t,\varphi}^{\text{HCSO}} \leq \pi_{m,t,\varphi}^{\text{HCSO}} \cdot P_{m,\max} \cdot I_{m,t,\varphi}^{\text{HCSO},*} \quad \forall m, t, \varphi \quad (44)$$

Due to upper level and lower level decision-makers are of opposite interest in our mathematic model, the second subproblem proposed in [39] is unnecessary, which is an important factor for adopting the CCG algorithm. Based on the above procedure, the original bi-level optimization problem is decomposed into two MILP problems. In this way, the proposed problem can be solved accurately and efficiently by commercial-off-the-shelf solvers such as Gurobi [40].

IV. CASE STUDY

A. Case Description and Basic Settings

In this case study, we consider a 7-node highway transportation network with an origin-destination (OD) pair, which is an

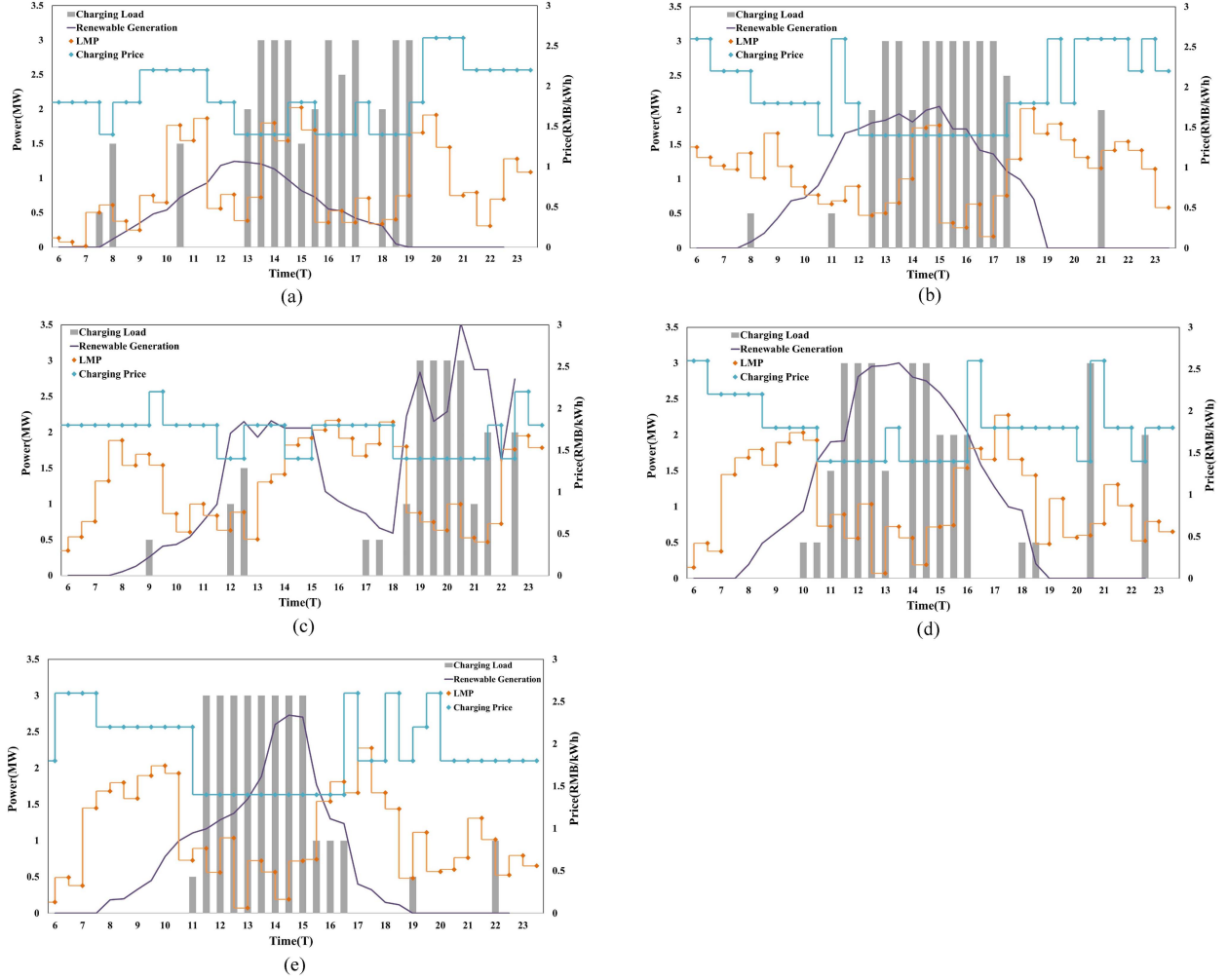


Fig. 6. Charging load, renewable generation, charging price and LMP profiles for each CS.

abstraction of the topological network based on a real-world highway system in China, as shown in Fig. 4. The total length of the traffic highway network is approximately 600 km, and the highway is evenly divided into six links. Five charging stations located in service zones are distributed during the highway, namely CS1, CS2, ..., and CS5. The origin and destination nodes are denoted as green and blue nodes, respectively. Due to the low traffic volume during that period, we ignore the EVs traveling on the highway between 0 a.m. and 6 a.m.. The operation horizon is discretized into 36-time intervals covering from 6:00 to 24:00.

Parameter settings are provided in Table I. Note that we consider four candidate pricing options for CSs. We assume that all CSs are installed with distributed PV resources and CS3 is equipped with distributed wind power generator resources. The installed capacity of each distributed resource is set to 3MW. Since the CSs are geographically distributed across the highway, the renewable generation profiles for each CS are different due to the solar radiation or other factors in their area, as shown in Fig. 5. For example, the area CS1 located is cloudy, while that for CS4 is well illuminated. We assume the departure time of vehicles follows a uniform distribution to mimic the actual situation. It shall be noted that there is no inflow of EVs from the

TABLE I
PARAMETERS SETTINGS OF EV AND CS

Parameters	Values
Battery capacity	75kWh
Energy consumption rate	0.15kWh/km
Average traveling speed	100km/h
Starting battery level	60kWh
Minimum battery capacity for each electric vehicle	7.5kWh
Maximal charging power limit for all charging station	3MW
Total length of highway	600km
Traffic flow	100EVs/hour
Charging power	30kW/h
Optional charging prices	1.4/1.8/2.2/2.6 RMB

charging station nodes as we assume the highway is operated in a closed manner. The traveling speed of EVs is assumed as 100 km/h based on the historical highway traveling data for EVs.

We design two cases to illustrate the effectiveness of the proposed pricing-based charging navigation scheme. We formulate the proposed pricing scheme as Base Case (i.e., charging price strategy varies in a spatial-temporal manner). In Comparison

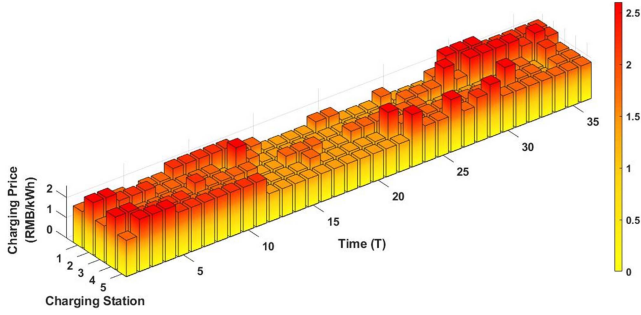


Fig. 7. Spatial-temporal distribution of charging prices in Base Case.

Case, time-of-use (TOU) price is adopted as the charging pricing strategy for all the CSs which only varies along with time.

The optimization model is implemented using MATLAB 2018b with Gurobi on a laptop with a 4-core Intel Core i7 processor and 8 GB memory.

B. Simulation Results

Fig. 6 illustrates the detailed spatial-temporal distribution of charging load at each CS. Meanwhile, it also contains renewable generation and LMP profiles to show the effectiveness of the strategy more distinctly. Firstly, under the proposed charging navigation scheme, the charging loads are more concentrated in the period when renewable energy is sufficient. Secondly, charging loads also tend to stay in the period when the LMPs are lower. Fig. 6 proves these two conclusions, the charging loads are mostly concentrated at 13:00-19:00 at CS 1, 13:00-17:00 at CS 2, 19:00-21:00 at CS 3, 11:00-16:00 at CS 4, and 11:00-15:00 at CS5. Among them, the charging loads of CS2, 3, and 4 are concentrated primarily due to the relatively high amount of renewable energy resources in these periods, while the optimization results for CS1 and CS5 illustrate that lower LMPs and renewable generation play a joint role in shaping the charging load. Thirdly, spatial-temporal charging prices shift the charging loads based on the navigation scheme. Taking Fig. 6(d) as an example, there is sufficient PV generation from 12:00 to 16:00 and the HCSO sets lower prices to attract more EVs to charge. In this way, the utilization of renewable generation can be enhanced. At 20:30 and 22:30, CS3 can purchase inexpensive electricity from the main grid, so it adopts a lower charging price strategy to navigate more EVs to charge. Fig. 7 shows the optimization results of spatial-temporal charging prices for each CS in Base Case. Note that since the profit cap limits the revenue of monopoly operators earning from EVs, each CS only sets a higher price during a few hours. Because the marginal cost for renewable generation is often negligible, in our framework, the HCSO optimizes its operation profit by utilizing more renewable energy via strategically determining the charging prices and reshapes the spatial-temporal distribution of charging loads.

We further illustrate the effectiveness of our proposed pricing-based charging navigation strategy for dispatching charging loads to dissipate clean energy by comparing the optimization results of Comparison Case, which are based on TOU prices.

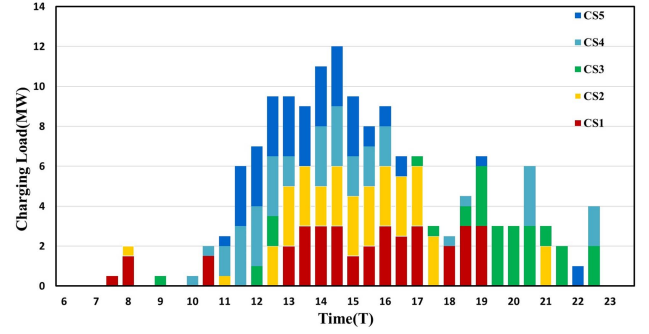


Fig. 8. Spatial-temporal distribution of charging loads in Base Case.

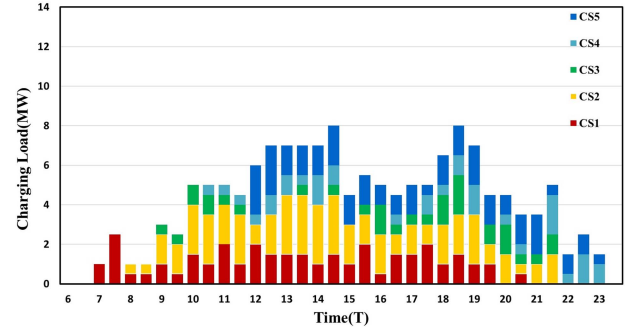


Fig. 9. Spatial-temporal distribution of charging loads in Comparison Case.

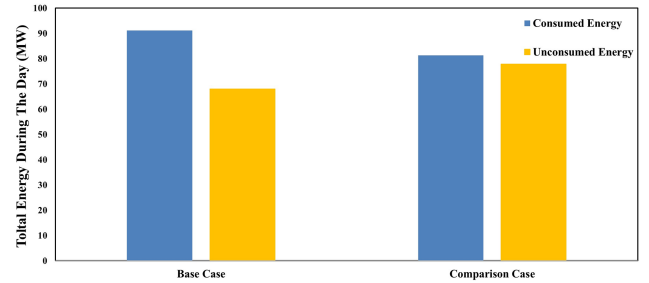


Fig. 10. Renewable energy utilization in two cases.

Figs. 8 and 9 show the spatial-temporal load distribution at each CS in Base Case and Comparison Case, respectively. Compared to the results in the Comparison Case, the peak loads in the Base Case are more prominent. Due to the different geographical locations of each CS, different prices at the same time can better navigate EVs to integrate renewable energy. 12:00-17:00 is the peak charging period because the areas tend to have more renewable energy sources this time. As a result, charging stations feature greater pricing flexibility by adopting spatial-temporal charging prices, routing EVs to charge at these locations with lower marginal costs.

Fig. 10 shows the renewable energy utilization of each CS in two cases. The detailed data is presented in Table II. In the Base Case where the spatial-temporal charging prices are adopted, EVs are routed by CSs properly and the renewable power integration could be promoted accordingly, and the utilization

TABLE II
SIMULATION RESULTS

Case	Renewable utilization rate	Power purchase cost
Base Case	57.28%	38774.91 RMB
Comparison Case	51.06%	61537.69 RMB

of renewable energy is increased by 6.22%. The electricity purchase cost in Base Case is 37.0% lower than that in Comparison Case. Due to the adoption of LMP, the spatial-temporal pricing strategy can set lower prices to attract EVs to charge at times where the LMPs are lower, thus reducing costs significantly. The experiment validated the economic benefit of EVs routing considering the spatial-temporal charging prices.

V. CONCLUSION

This paper presents a pricing-based charging navigation scheme for cross-regional charging station operators in the context of highway transportation. Meanwhile, distributed renewable energy resources are accounted in the optimization framework. In addition, a time-space network model with highway characteristics is developed to portray the spatial-temporal coupling characteristics of charging planning. We model the charging price-setting scheme as a bi-level optimization model, where the HCSO maximizes its own profit, and the EVs choose the least costly traveling option. It is reformulated as an intricate single-level optimization problem, with upper and lower bounds dynamically generated by a column generation algorithm that iterates continuously to reach the optimal solution. Our case study demonstrates the effectiveness of this approach, where charging station operators dynamically optimize pricing to navigate EVs charging to dissipate renewable energy while enabling spatial-temporal load shifting.

As for the future directions for this work, we can adopt a stochastic formulation to better depict the uncertainties in the pricing navigation process. Also, data-driven and machine learning based methods can be investigated to improve the solution efficiency for the proposed bi-level model.

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